BULD EXCHANGE: AI POWERED CONSTRUCTION MATERIAL EXCHANGE PLATFORM

MINI PROJECT REPORT

Submitted by

ALEX ANTONY

CHN25MCA-2012

to

APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of Degree in

Master of Computer Applications



DEPARTMENT OF COMPUTER ENGINEERING

COLLEGE OF ENGINEERING CHENGANNUR, ALAPPUZHA

OCTOBER 2026

DEPARTMENT OF COMPUTER ENGINEERING
COLLEGE OF ENGINEERING CHENGANNUR
ALAPPUZHA

CERTIFICATE

This is to certify that the project report titled ***BULD EXCHANGE: AI POWERED CONSTRUCTION MATERIAL EXCHANGE PLATFORM*** *is a bonafide record of the* ***20MCA245 MINI PROJECT*** *presented by* ***ALEX ANTONY*** *(CHN25MCA-2012), Third Semester Master of Computer Applications student, under our guidance and supervision. This project is submitted in partial fulfillment of the requirements for the award of the degree* ***Master of Computer Applications*** *of* ***APJ Abdul Kalam Technological University****.*

Smt. Alka Vijay
Project Guide
Assistant Professor
Dept. of Computer Engineering

Smt. Josmi Jose
Project Coordinator
Assistant Professor
Dept. of Computer Engineering

Dr. Renu George
Professor
Head of the Department
Dept. of Computer Engineering

DECLARATION

I undersigned hereby declare that the project report "**BULD EXCHANGE: AI POWERED CONSTRUCTION MATERIAL EXCHANGE PLATFORM**", submitted for partial fulfillment of the requirements for the award of degree of Master of Computer Application of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under the supervision of **Smt. Alka Vijay**, Assistant Professor, Department of Computer Engineering. This submission represents my ideas in my own words, and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources. I also declare that I have adhered to the ethics of academic honesty and integrity and have not misrepresented or fabricated any data, idea, fact, or source in this submission. I understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma, or similar title of any other University.

Place: Chengannur **ALEX ANTONY**

Date : **CHN25MCA-2012**

ACKNOWLEDGEMENT

This work would not have been possible without the support of many people. First and foremost, I give thanks to Almighty God who gave me the inner strength, resources, and ability to complete my project successfully.

I would like to thank **Dr. Hari V.S** , The Principal , College of Engineering Chengannur, for providing the best facilities and atmosphere for the project completion and presentation. Special thanks to **Dr. Renu George** , HOD , Department of Computer Engineering, for her exceptional support, guidance, and encouragement throughout the project. I would also like to thank my project coordinators **Smt. Josmi Jose** , Assistant Professor, Department of Computer Engineering , **Smt. Santhy Viswam** , Assistant Professor, Department of Computer Engineering ,**Smt. Celin T** , Assistant Professor, Department of Computer Engineering , for their extended help and support during the project. I would also like to thank **Smt. Alka Vijay** , Assistant Professor, Department of Computer Engineering, who also took on the role of my project guide , for her extended help and support during the project.

I would like to thank my dear friends and faculty for extending their cooperation and encouragement throughout the project work, without which I would never have completed the project this well. Thank you all for your love and also for being very understanding.

ALEX ANTONY

CHN25MCA-2012

ABSTRACT

BuildExchange is a MERN-stack web application designed to promote the reuse of surplus construction materials such as bricks, cement, tiles, steel, wood, and pipes. It provides a centralized platform where sellers can list available materials with details such as category, condition, price, quantity, location, and images, while buyers can browse, search, filter, add materials to favorites, and submit exchange requests. The system supports request management, messaging, notifications, and pickup coordination between buyers and sellers. Material quantities are updated automatically after collection to maintain accurate inventory records. The platform also incorporates perceptual hashing (pHash) to identify potentially duplicate or visually similar listing images for administrator review. MongoDB is used to manage application data, while Cloudinary provides cloud-based image storage. Role-based access control supports separate functionalities for buyers, sellers, and administrators. BuildExchange aims to reduce construction waste, improve resource utilization, and help users find affordable construction materials. By combining a digital marketplace with image similarity analysis and organized material collection, the system encourages sustainable construction practices and supports a more efficient reuse of available resources.

CONTENTS

Declaration i

Acknowledgement ii

Abstract iii

List of Figures vi

List of Tables vii

1 INTRODUCTION 1

1.1 Project Area 1

1.2 Objectives 2

2 LITERATURE REVIEW 3

2.1 WasteSAM: A Novel Approach for Accurate Construction Waste Image Segmentation (2025) 3

2.2 D5 Digital Circular Workflow 3

2.3 The Butterfly Matchmaking Model for Circular Construction 4

2.4 Building Material Reuse: An Optimization Framework 4

3 PROBLEM DEFINITION AND MOTIVATIONS 5

3.1 Existing System 5

3.2 Limitations 5

3.3 Problem Statement 6

3.4 Proposed System 6

4 REQUIREMENT ANALYSIS 9

4.1 Hardware Requirements 9

4.2 Software Requirements 9

4.3 System Features 10

4.4 Non-Functional Requirements 11

4.4.1 Performance Requirements 11

4.4.2 Quality Requirements....11

5 DESIGN AND IMPLEMENTATION 12

5.1 Overall Design....12

5.2 System Design....12

5.2.1 Server-Side Design....13

5.2.2 Client-Side Design....14

5.3 System Architecture....14

5.4 Use Case Diagram....16

5.5 Data Flow Diagrams (DFD)....17

5.6 Module Description....20

5.6.1 Admin Module....20

5.6.2 Seller Module....20

5.6.3 Buyer Module....21

6 SYSTEM IMPLEMENTATION 22

6.1 Implementation Plan....22

6.1.1 Setup and Environment Preparation....22

6.1.2 Backend Implementation....22

6.1.3 Frontend Implementation....23

6.1.4 Security and Data Handling....23

6.2 Testing and Various Types of Testing Used....24

6.2.1 Unit Testing....24

6.2.2 Integration Testing....25

6.2.3 System Testing....25

7 RESULTS AND DISCUSSION 27

7.1 Advantages....27

7.2 Limitations....28

7.3 Screenshots....29

8 CONCLUSION AND FUTURE SCOPE 35

8.1 Conclusion....35

8.2 Future Scope....35

REFERENCES 37

LIST OF FIGURES

5.1 System Architecture 15

5.2 Use Case Diagram..... 16

5.3 Data Flow Diagram level 0..... 18

5.4 Data Flow Diagram level 1-Admin..... 18

5.5 Data Flow Diagram level 1-Seller..... 19

5.6 Data Flow Diagram level 1-Buyer..... 19

7.1 Home Page 29

7.2 Registration Page 29

7.3 Login Page 29

7.4 Admin Dashboard 30

7.5 Admin Material Management 30

7.6 Admin User Management 30

7.7 AI Image Monitoring 31

7.8 Seller Dashboard 31

7.9 Seller Order 31

7.10 Seller Request 32

7.11 Image Verification 32

7.12 Seller History 32

7.13 Buyer Dashboard 33

7.14 Browse Materials 33

7.15 Buyer Order 33

7.16 Favorite 34

7.17 Chat 34

LIST OF TABLES

6.1 Unit Test Cases and Results 24
6.2 Integration Test Cases and Results 25
6.3 System Test Cases and Results 25

CHAPTER 1

INTRODUCTION

1.1 Project Area

The construction industry plays a significant role in economic development and infrastructure growth. However, construction activities often generate surplus and unused materials such as bricks, cement, tiles, steel, wood, and pipes. These materials may be discarded or left unused due to poor resource management and difficulties in connecting sellers with potential buyers. This results in material wastage, increased construction costs, and environmental concerns. The effective reuse of surplus construction materials can help reduce waste, improve resource utilization, and promote sustainable construction practices.

Build Exchange: AI-Powered Construction Material Exchange Platform is a web-based application developed to provide a centralized digital marketplace for exchanging surplus construction materials. The platform connects sellers who have unused materials with buyers who require them. Sellers can list materials by providing details such as material name, category, condition, price, quantity, location, and images. Buyers can browse available listings, search and filter materials ac-cording to their requirements, add materials to their favorites, and submit requests for the required quantities.

The application supports request management, messaging, notifications, and pickup scheduling to improve communication and coordination between buyers and sellers. It also provides automatic inventory updates when a collection is marked as completed, ensuring that the available material quantity remains accurate. AI-assisted image analysis is incorporated to identify duplicate or visu-ally similar listing images and flag potentially suspicious content for administrative review. Cloud-inary is used for cloud-based image storage and delivery, while MongoDB manages user details, material listings, requests, and other application data.

Build Exchange is a MERN stack web application (MongoDB, Express.js, React.js, Node.js) for exchanging surplus construction materials. By integrating digital marketplace features, automated inventory tracking, and AI-assisted image verification via Cloudinary, the platform improves mate-rial accessibility, reduces construction waste, and promotes sustainable resource management.

1.2 Objectives

1. To develop a centralized web-based platform that connects buyers and sellers of surplus and unused construction materials, facilitating the exchange and reuse of materials.
2. To provide an efficient material listing and management system that allows sellers to add, update, and manage construction materials with details such as category, condition, price, quantity, location, and images.
3. To implement search and filtering facilities that enable buyers to find suitable construction materials based on material name, category, price, condition, quantity, and location.
4. To develop a request management system that allows buyers to request required materials and quantities while enabling sellers to accept or reject requests based on availability.
5. To facilitate communication and pickup scheduling between buyers and sellers through messaging, notifications, and the sharing of pickup locations, dates, and times.
6. To maintain accurate inventory records by automatically updating the available material quantity after a collection is completed and preventing requests for unavailable materials.
7. To incorporate AI-assisted image analysis using techniques such as perceptual hashing (pHash) to identify duplicate or visually similar listing images and flag potentially suspicious content for administrative review.
8. To integrate Cloudinary for cloud-based storage and delivery of material images while maintaining the corresponding image URLs and related information in MongoDB.
9. To implement role-based access control for buyers, sellers, and administrators to ensure that each user can access the appropriate features and management functions.
10. To promote sustainable construction practices by encouraging the reuse of surplus materials, reducing unnecessary disposal, and improving the efficient utilization of construction resources.
11. To develop a scalable and maintainable web application using the MERN stack (MongoDB, Express.js, React.js, and Node.js) that provides a responsive user interface and supports future enhancements.

CHAPTER 2

LITERATURE REVIEW

2.1 WasteSAM: A Novel Approach for Accurate Construction Waste Image Segmentation

Seokjae Heo and Seunguk Na introduced Developing WasteSAM: A Novel Approach for Ac-curate Construction Waste Image Segmentation to Facilitate Efficient Recycling. The study presents WasteSAM, an enhanced version of the Segment Anything Model (SAM) adapted to construction waste imagery. The researchers used a dataset containing more than 15,000 masks representing five categories of construction materials. The proposed method improved segmentation performance compared with the original SAM model, reporting an average improvement of 23.9 in the Dice similarity coefficient and 30.0 in normalized surface distance metrics. The study demonstrates how segmentation models can help identify the regions occupied by waste materials within images. However, the quality and diversity of training data remain important for reliable performance in real-world environments. This research is relevant to the proposed platform because image segmen-tation techniques may support more detailed analysis of construction material photographs. [5]

2.1 D5 Digital Circular Workflow

Catherine De Wolf and colleagues proposed the D5 Digital Circular Workflow: Five Digital Steps Towards Matchmaking for Material Reuse in Construction. The study presents a digital work-flow designed to integrate technologies into construction material reuse. The workflow consists of five stages: Detection, Disassembly, Distribution, Design and Deployment. It explores the use of machine learning, computer vision, building information modelling (BIM), reality capture and dig-ital product passports to identify, document, track and reuse construction materials. The proposed workflow was examined through real-world case studies, demonstrating how digital technologies can support material reuse throughout the construction process. However, coordinating material characteristics, condition, design requirements and transportation remains challenging. This study provides a foundation for the proposed platform's digital material management and AI-assisted analysis features. [1]

2.2 The Butterfly Matchmaking Model for Circular Construction

The study The Butterfly Matchmaking Model for Circular Construction: Towards a Digital Matchmaking Platform Tailored to French Policy investigates how digital platforms can improve the connection between the supply and demand of reusable construction materials. The researchers used participatory action research to understand the requirements of construction-sector stakeholders. They proposed two interconnected matchmaking loops: one supporting deconstruction activities and another supporting construction activities. The model emphasizes information exchange and coordination among contractors, collectors, designers and material dealers. Its findings highlight the importance of designing digital platforms around stakeholder requirements rather than simply listing available materials. However, the model was developed in the French context and requires further practical testing and validation. This research supports the proposed platform's buyer–seller interaction, material-request workflow and matching of available materials with buyer requirements.[2]

2.3 Building Material Reuse: An Optimization Framework

Adama Olumo and Carl Haas presented Building Material Reuse: An Optimization Framework for Sourcing New and Reclaimed Building Materials. The study addresses the difficulties associated with sourcing reclaimed construction materials, which is often a manual process affected by limited digital availability of material information. The authors developed a modular framework consisting of three components: the Real Environment, which uses tools such as 3D scanners and spreadsheets; the Model Environment, which incorporates Building Information Modelling and Life Cycle Assessment; and the Core Engine, which performs optimization. The framework was evaluated using real-world material data and a building case study. The research demonstrates how optimization can support the selection of reclaimed and new materials based on project requirements. However, the selection process may need to consider additional factors, including material condition, mechanical properties and structural suitability. This research is relevant to the proposed platform because it highlights the importance of organizing material information and helping users find suitable reusable materials.[3]

CHAPTER 3

PROBLEM DEFINITION AND MOTIVATIONS

3.1 Existing System

The existing system for construction material exchange mainly relies on traditional methods such as local dealers, personal contacts, phone calls, and general online marketplaces. Sellers often face difficulties finding buyers for surplus construction materials, while buyers may need to contact multiple sellers to find suitable materials. Material availability, price, quantity, and condition are frequently communicated manually, making the process time-consuming and difficult to manage.

Although online marketplaces provide facilities for listing and searching products, they may not support the complete construction material exchange workflow, including request management, inventory updates, pickup scheduling, and AI-assisted image verification. These limitations highlight the need for a dedicated platform to simplify material exchange, reduce construction waste, and promote the reuse of surplus materials.

3.2 Limitations

- **Fragmented Information :** : Details about available surplus construction materials may be distributed across multiple sellers and platforms, making them difficult to access centrally.
- **Time-Consuming Search :** Buyers may need to contact several sellers or visit different locations to find appropriate materials.
- **Inaccurate Inventory Information :** Manual updates may result in outdated material quantities and availability details.
- **Manual Request Management :** Requests and confirmations may depend on separate communication channels, making progress difficult to track.
- **Pickup Coordination Difficulties :** Arranging collection locations, dates, and times through manual communication may cause delays or misunderstandings.
- **Limited Image Verification :** Platforms without suitable image-analysis tools may not automatically identify duplicate or visually similar listing images.

- **Limited Integrated Administration :** Some platforms may lack centralized facilities for managing users, categories, complaints, listings, and flagged content.
- **Challenges in promoting material reuse :** When surplus materials are difficult to discover or compare, potentially reusable resources may remain unused or be discarded instead of being made available to other users.
- **Difficulty Tracking Exchange Progress :** Without structured status tracking, buyers and sellers may find it difficult to determine whether a request is pending, accepted, scheduled for pickup, or completed.
- **Limited Role-Based Access Control :** Systems without clearly defined user permissions may not adequately restrict administrative functions or protect operations that should only be performed by authorized users recommending prices.
- **Duplicate Material Listings :** The same material may be listed multiple times, creating confusion for buyers and making it difficult to identify the actual availability of materials.

3.3 Problem Statement

To develop an AI-powered web platform that connects buyers and sellers, promotes material reuse, and helps reduce construction waste.

3.4 Proposed System

The proposed system, Build Exchange: AI-Powered Construction Material Exchange Platform, is a web-based application designed to simplify the exchange of surplus and unused construction materials between buyers and sellers. The system provides a centralized platform where users can list, discover, request, and manage construction materials efficiently. It aims to overcome the limitations of traditional material exchange methods by integrating material management, request processing, inventory control, communication, pickup scheduling, and AI-assisted image verification into a single application.

The platform allows sellers to register and publish material listings by providing details such as material name, category, description, condition, expected price, available quantity, location, and images. Buyers can browse available materials, search and filter listings according to their requirements, compare material details, add items to their favorites, and submit requests for the required quantities. Sellers can review incoming requests and accept or reject them based on material availability.

After a request is accepted, the system facilitates communication between the buyer and seller to coordinate the collection process. The seller can share the pickup location, date, time, and additional instructions through the platform. The system maintains request and collection statuses so that both parties can monitor progress. Once the collection is completed, the available material quantity is automatically reduced by the collected amount. When the remaining quantity reaches zero, the listing is marked as sold or unavailable to prevent further requests.

The proposed system incorporates AI-assisted image verification** to identify duplicate or visually similar images uploaded with material listings. Techniques such as perceptual hashing (pHash) can be used to compare images and flag potential duplicates for administrative review. This feature helps improve listing quality and supports the identification of repeated or potentially suspicious image submissions. Image similarity is treated as an indicator for review rather than definitive proof of fraudulent activity.

The system uses Cloudinary for cloud-based storage and delivery of material images, while MongoDB stores user details, material information, requests, and other application records. The application is developed using the MERN stack, consisting of MongoDB, Express.js, React.js, and Node.js. Role-based access control provides separate functionalities for buyers, sellers, and administrators. Administrators can manage users, material categories, listings, complaints, reports, and flagged images to maintain the overall quality and reliability of the platform.

The proposed system does not include online payment processing; instead, it focuses on material discovery, request management, and coordination of material collection. By providing an organized and accessible digital marketplace, Build Exchange aims to reduce unnecessary construction material waste, improve the availability of reusable resources, and encourage sustainable construction practices.

1. Centralized Platform: Provides a single platform for buyers and sellers to exchange surplus construction materials.
2. Efficient Material Discovery: Enables buyers to search and filter materials according to category, price, condition, quantity, and location.
3. Simplified Listing Management: Allows sellers to create, update, and manage material listings conveniently.
4. Automated Inventory Updates: Automatically adjusts available material quantities after completed collections and prevents requests exceeding available stock.

5. Improved Request Tracking: Maintains request statuses and helps buyers and sellers monitor the exchange process.
6. Convenient Pickup Coordination: Supports communication and scheduling of material collection through the platform.
7. AI-Assisted Image Verification: Helps identify duplicate or visually similar listing images for further review.
8. Cloud-Based Image Storage: Uses Cloudinary to store and deliver images efficiently.
9. Role-Based Access Control: Provides appropriate permissions for buyers, sellers, and administrators.
10. Reduced Material Waste: Encourages the reuse of surplus construction materials and supports more sustainable resource management.

CHAPTER 4

REQUIREMENT ANALYSIS

4.1 Hardware Requirements

- **Processor :** Intel Core i3 or equivalent
- **RAM :** Minimum 4 GB
- **Storage :** Minimum 10 GB free space
- **Display :** 1366 × 768 resolution
- **Internet Connection :** Stable connection
- **Input Devices :** Keyboard and mouse

4.2 Software Requirements

- **Operating System :** Windows 10/11 or Linux
- **Frontend :** React.js
- **Programming Languages :** JavaScript, HTML, CSS
- **Backend :** Node.js, Express.js
- **Database :** MongoDB
- **Image Storage :** Cloudinary
- **AI Image Processing :** Perceptual Hashing (pHash)
- **Code Editor :** Visual Studio Code
- **API Testing :** Postman
- **Package Manager :** npm
- **Version Control :** Git and GitHub

4.3 System Features

- **User Registration:** Allows students to create accounts by providing their personal and academic details, with passwords stored securely using hashing.
- **User Login:** Authenticates registered students using their email and password to access marketplace features.
- **User Profile:** Displays students personal and academic information, including basic details useful for identifying sellers.
- **Add Product:** Allows students to list products by entering product details, price, and images.
- **AI Product Description Generation:** Uses AI to generate clear and informative product descriptions from the details provided by sellers.
- **AI Price Recommendation:** Uses AI to suggest a suitable selling price based on available product information.
- **Marketplace:** Displays listed products with their images, names, categories, descriptions, and prices for students to browse.
- **Product Search:** Allows users to find products using product names or relevant keywords through conventional database searching.
- **Seller Information:** Displays relevant seller details so interested buyers can contact sellers about products.
- **My Products:** Allows sellers to view and manage the products they have listed.
- **Edit Product:** Enables sellers to update existing product details, including the name, description, category, price, and image.
- **Admin Login:** Provides a separate authentication facility for administrators to access management functions securely.
- **Admin Product Management:** Allows administrators to view and manage product listings on the platform.
- **Logout:** Allows students and administrators to securely end their sessions.
- **Responsive User Interface:** Provides a user-friendly interface that adapts to desktops, laptops, tablets, and smartphones using Bootstrap.
- **Admin User Management:** Enables administrators to view and manage registered student accounts.

4.4 Non-Functional Requirements

4.4.1 Performance Requirements

1. **Fast Response:** The system should respond to user requests quickly without unnecessary delays.
2. **Efficient Database Access:** The system should retrieve and store data efficiently in the MySQL database.
3. **Fast Product Search:** Product-name searches should display relevant results quickly.
4. **Image Handling:** Product images should be uploaded and displayed without significantly slowing down the system.
5. **AI Response:** AI features should generate descriptions and price recommendations within a reasonable time.
6. **Multiple Users:** The system should support multiple students using the application at the same time.

4.4.2 Quality Requirements

1. Usability: The system should provide a simple and user-friendly interface that users can easily understand and navigate.
2. Reliability: The application should perform its functions consistently with minimal errors and interruptions.
3. Security: User accounts and stored information should be protected from unauthorized access through appropriate authentication and access control.
4. Maintainability: The application should be easy to maintain, update, and modify when required.
5. Scalability: The system should support the addition of more users, material listings, and features in the future.
6. Availability: The application should be accessible to users whenever the server and required services are operational.
7. Accuracy and Data Integrity: User information, material details, quantities, and exchange requests should be stored, processed, and displayed accurately and consistently.
8. Compatibility: The application should function properly on commonly used modern web browsers and different screen sizes.

CHAPTER 5

DESIGN AND IMPLEMENTATION

5.1 Overall Design

Build Exchange: AI-Powered Construction Material Exchange Platform is designed as a web-based application that connects buyers and sellers of surplus construction materials through a centralized platform. The system follows a client–server architecture, where the client-side interface communicates with the server-side application through REST APIs. MongoDB is used to store application data, while Cloudinary provides cloud-based storage and delivery of material images.

The overall design consists of three main user roles: Buyer, Seller, and Administrator. Buyers can browse materials, search and filter listings, add materials to favorites, and submit material requests. Sellers can create and manage listings, respond to requests, coordinate material collection, and update availability. Administrators can manage users, categories, listings, complaints, reports, and images flagged for review.

The system also incorporates AI-assisted image analysis to identify duplicate or visually similar listing images. These features help improve listing management and support the administrative review process. The overall design emphasizes usability, security, maintainability, and efficient material exchange.

5.2 System Design

The system design defines the major components of the application and their interactions. Build Exchange is organized into the following components:

- Presentation Layer: Provides the user interface for buyers, sellers, and administrators.
- Application Layer: Handles business logic, authentication, material management, requests, inventory updates, and communication.
- Database Layer: Stores user information, material listings, categories, requests, and other application records in MongoDB.
- Image Storage Layer: Uses Cloudinary to store and deliver material images.

- AI Image Analysis Module: Processes listing images to identify potential duplicates or visually similar images and flags matches for review.

The client communicates with the backend through REST APIs. The backend validates incoming requests, checks user permissions, performs the required operations, and interacts with MongoDB and Cloudinary as needed. This modular structure makes the application easier to maintain, test, and extend.

5.2.1 Server-Side Design

The server side is developed using Node.js and Express.js. It handles application logic, processes client requests, validates submitted data, enforces authorization rules, and communicates with MongoDB and Cloudinary.

The backend is organized into routes, controllers, models, middleware, and supporting services. Routes define the API endpoints, controllers process requests, models manage database operations through Mongoose, and middleware handles functions such as authentication and access control.

The main server-side functions include:

1. Authentication and Authorization: Verifies user identity and restricts access according to assigned roles.
2. User Management: Handles user registration, login, profile operations, and administrative user management.
3. Material Management: Creates, retrieves, updates, and manages construction material listings.
4. Request Management: Processes buyer requests and seller acceptance or rejection.
5. Inventory Management: Validates available stock and updates quantities after completed collections.
6. Image Management: Handles image uploads and retrieval using Cloudinary, subject to the configured storage implementation.
7. AI Image Analysis: Compares listing images to identify potential duplicates or visually similar content.
8. Communication Management: Supports messaging, notifications, and pickup coordination.
9. Administrative Management: Supports listing moderation, complaint handling, category management, and review of flagged images.

5.2.2 Client-Side Design

The client side of Build Exchange is developed using React.js, HTML, CSS, and JavaScript. It provides an interactive interface through which users access the platform's features. The interface is organized according to user roles, ensuring that buyers, sellers, and administrators can access their respective functions.

The buyer interface provides material browsing, search and filtering, material details, favorites, request submission, request tracking, messaging, and pickup information. The seller interface supports material listing creation, listing management, incoming request handling, pickup scheduling, and inventory status updates. The administrator interface provides facilities for managing users, categories, material listings, complaints, reports, and flagged images.

Reusable React components are used to maintain consistency across pages and simplify interface development. Forms are used to collect material details and user inputs, while validation helps prevent incomplete or invalid submissions. The client communicates with the backend using HTTP requests to retrieve data and submit user actions. The interface displays relevant messages and status updates based on the server responses.

5.3 System Architecture

Build Exchange is a client–server platform built on the MERN stack (MongoDB, Express.js, React.js, and Node.js) that features an interactive React.js, HTML, CSS, and JavaScript interface. The system supports three primary user roles: buyers who can browse, manage favorites, and submit exchange requests; sellers who manage listings and coordinate pickups; and administrators who oversee users, listings, reports, and complaints. Additionally, an AI image analysis module utilizes perceptual hashing (pHash) to automatically detect duplicates and flag similar images for administrative review.

The server-side application uses Node.js and Express.js to handle client requests through REST APIs. It manages user authentication, role-based authorization, material listings, exchange requests, inventory management, and other application operations.

MongoDB serves as the database for storing user information, material details, categories, favorites, and exchange request records. Cloudinary is used for cloud-based storage and delivery of material images. The application also includes an image analysis module based on perceptual hashing (pHash), which compares uploaded images to identify exact or visually similar images and flag potential duplicates for review.

The client communicates with the server through API requests and responses. The server interacts with the database, image storage service, and image analysis functionality as required. This modular architecture supports maintainability, data management, and the efficient coordination of material exchanges between buyers and sellers. Build Exchange facilitates material requests and pickup coordination without processing online payments.

The system operates through the following workflow:

1. The user accesses the application through a web browser.
2. The React frontend displays the relevant interface based on the user's role.
3. The frontend sends requests to the Express.js backend through REST APIs.
4. MongoDB stores or retrieves application records.
5. Cloudinary stores or delivers material images when cloud image storage is used.
6. The AI image-analysis module checks listing images for potential duplicates or visual similarities.
7. The backend returns the operation result to the frontend, which updates the displayed information.

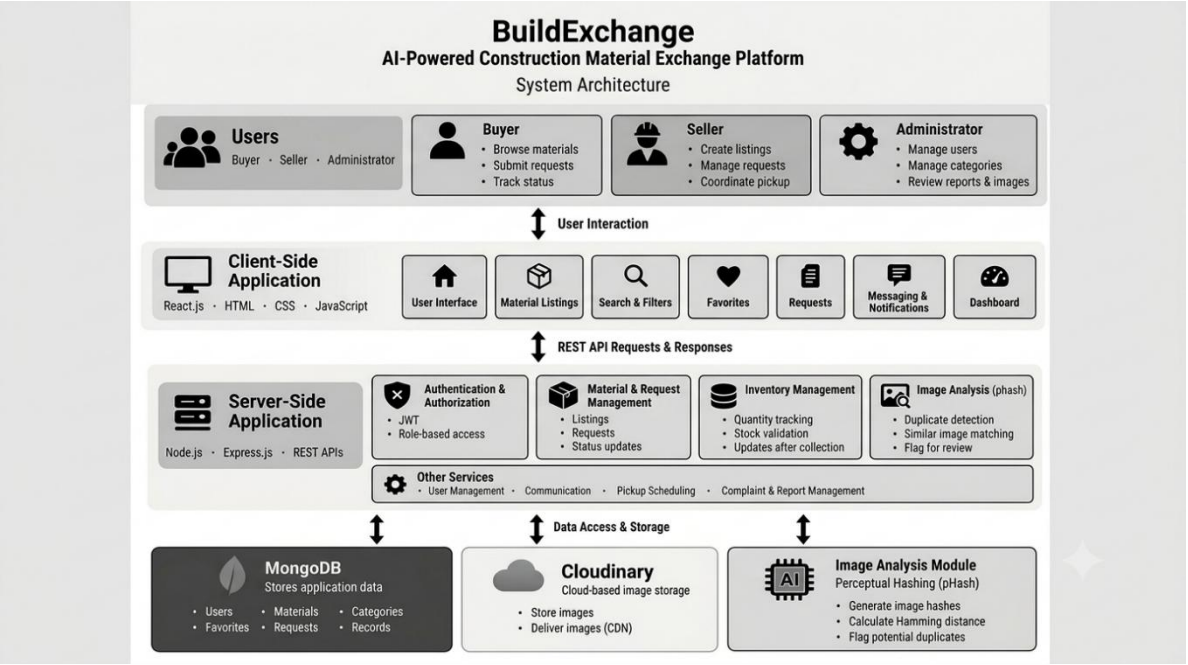


Figure 5.1: System Architecture

5.4 Use Case Diagram

The use case diagram represents the interactions between the three main actors — Buyer, Seller, and Administrator — and the functionalities provided by the Build Exchange construction material exchange platform.

The diagram includes material browsing, listing management, exchange requests, pickup coordination, image verification, and administrative management. Online payment functionality is excluded because the system does not process monetary transactions.

- Buyer: Register/login, browse and filter listings, manage favorites, and submit/track exchange requests.
- Seller: Manage listings, respond to requests, coordinate pickups, and update material quantities post-collection.
- Administrator: Manage users, categories, and listings, view reports, handle complaints, and review flagged images.
- AI Image Analysis: Uses perceptual hashing (pHash) to detect duplicates and flag similar images for review.

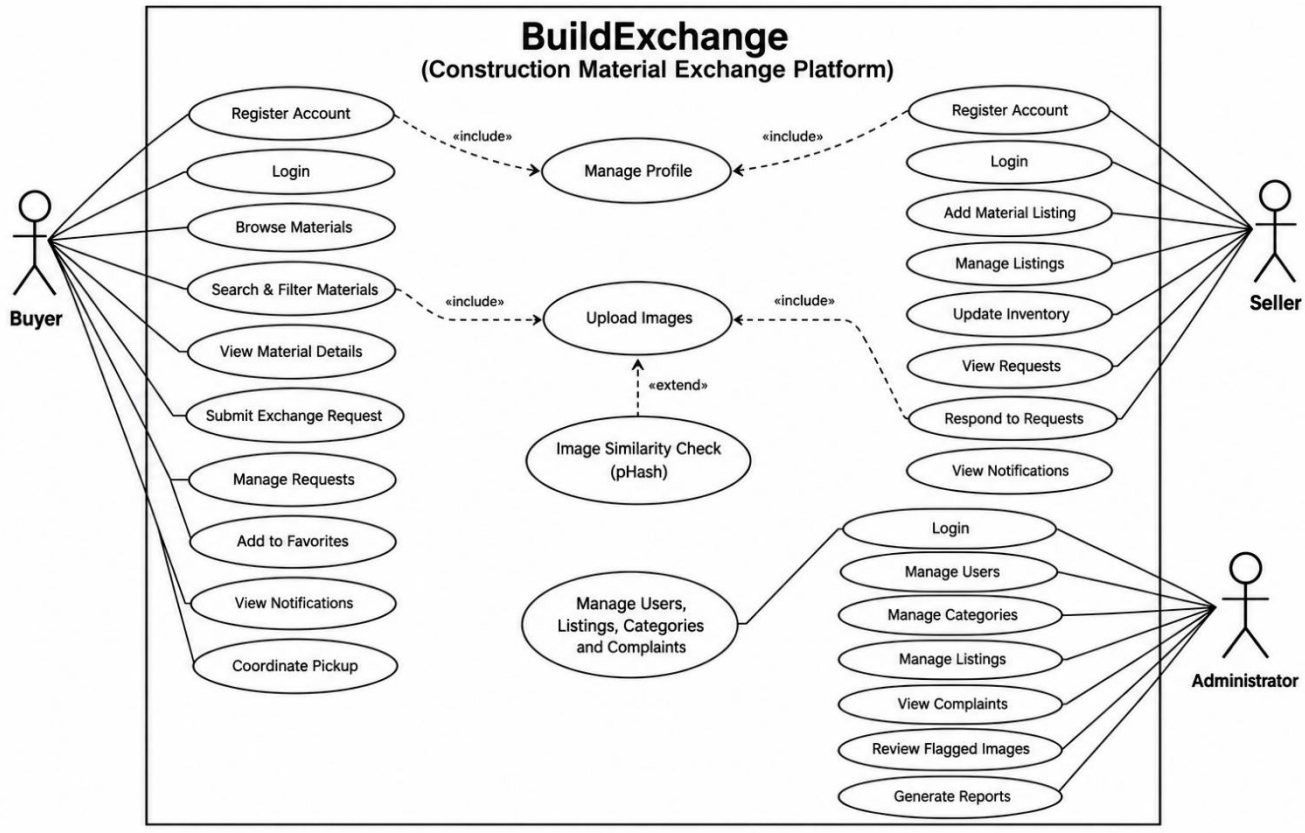


Figure 5.2: Use Case Diagram

5.5 Data Flow Diagrams (DFD)

The Data Flow Diagram (DFD) of Build Exchange illustrates how data flows between users, system processes, and storage components. It explains how information is received, processed, stored, updated, and retrieved within the application. The DFD represents the interactions between buyers, sellers, and administrators and shows how the platform manages material listings, exchange requests, inventory updates, and image analysis.

Level 0 DFD: The Level 0 DFD represents the entire Build Exchange platform as a single process, with three main external entities: Buyer, Seller, and Administrator. Buyers browse available construction materials, view details, and submit exchange requests. Sellers create and manage material listings, upload images, and respond to buyer requests. Administrators manage users, categories, listings, complaints, and reports. The system processes these inputs and provides the required information to each user. It also uses MongoDB for application data storage, Cloudinary for image storage, and a pHash-based image analysis module to identify potentially duplicate or visually similar images for review.

Level 1 DFD: The Level 1 DFD divides the system into five major processes. **User Management** handles registration, login, profiles, and role-based access. **Material Management** manages material details, categories, prices, quantities, locations, and listing availability. **Request and Order Management** processes buyer requests, seller responses, request status, messaging, and pickup coordination. It also updates material quantities after collection is completed. **Admin and Report Management** supports user and listing administration, complaint handling, and report generation. **Image Processing and Analysis** uses perceptual hashing (pHash) to compare uploaded images with existing images and flag potential duplicates or visually similar images for administrator review. This analysis assists image verification but does not independently confirm whether an image is fake or AI-generated.

Data Storage and Conclusion: MongoDB stores user details, material information, categories, favorites, exchange requests, and relevant image analysis records, while Cloudinary stores and delivers material images. These components work with the frontend and backend to maintain consistent data flow throughout the application. Overall, the DFD provides a clear view of the system's data-processing activities and demonstrates how Build Exchange supports material listing, exchange request management, inventory control, and image similarity analysis. The platform promotes construction material reuse, improves coordination between buyers and sellers, and helps reduce construction waste without processing online payments.

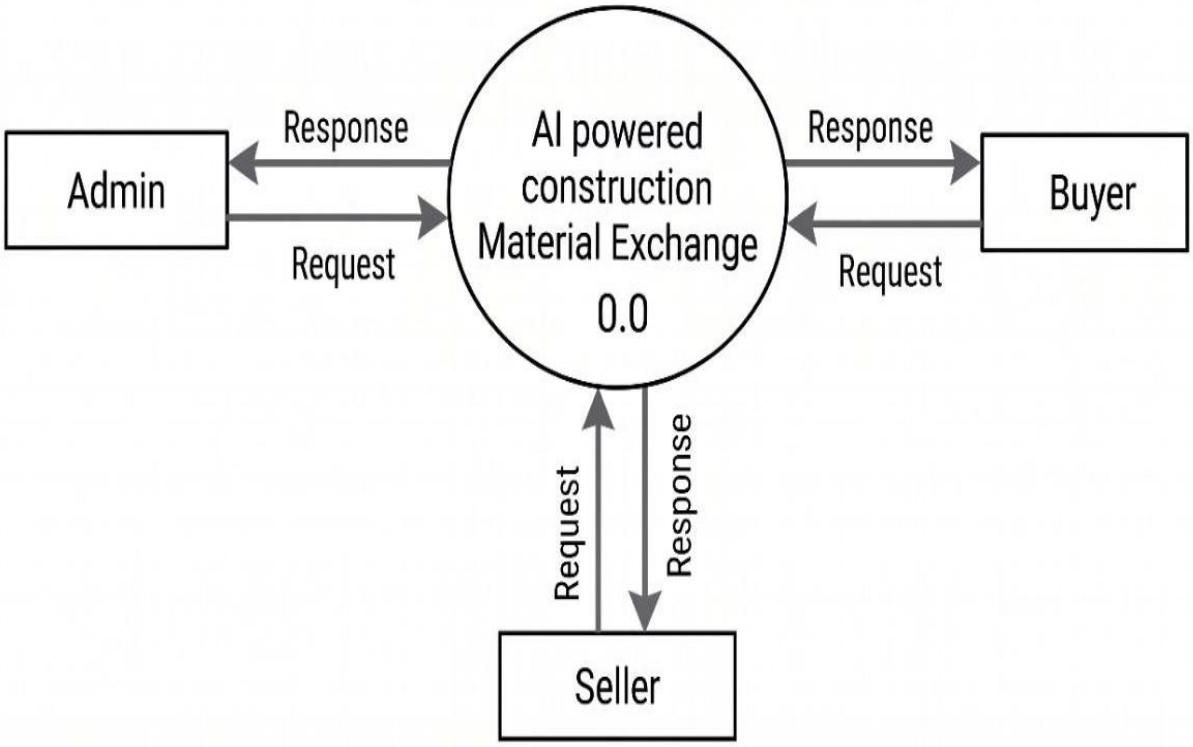


Figure 5.3: DFD Level 0

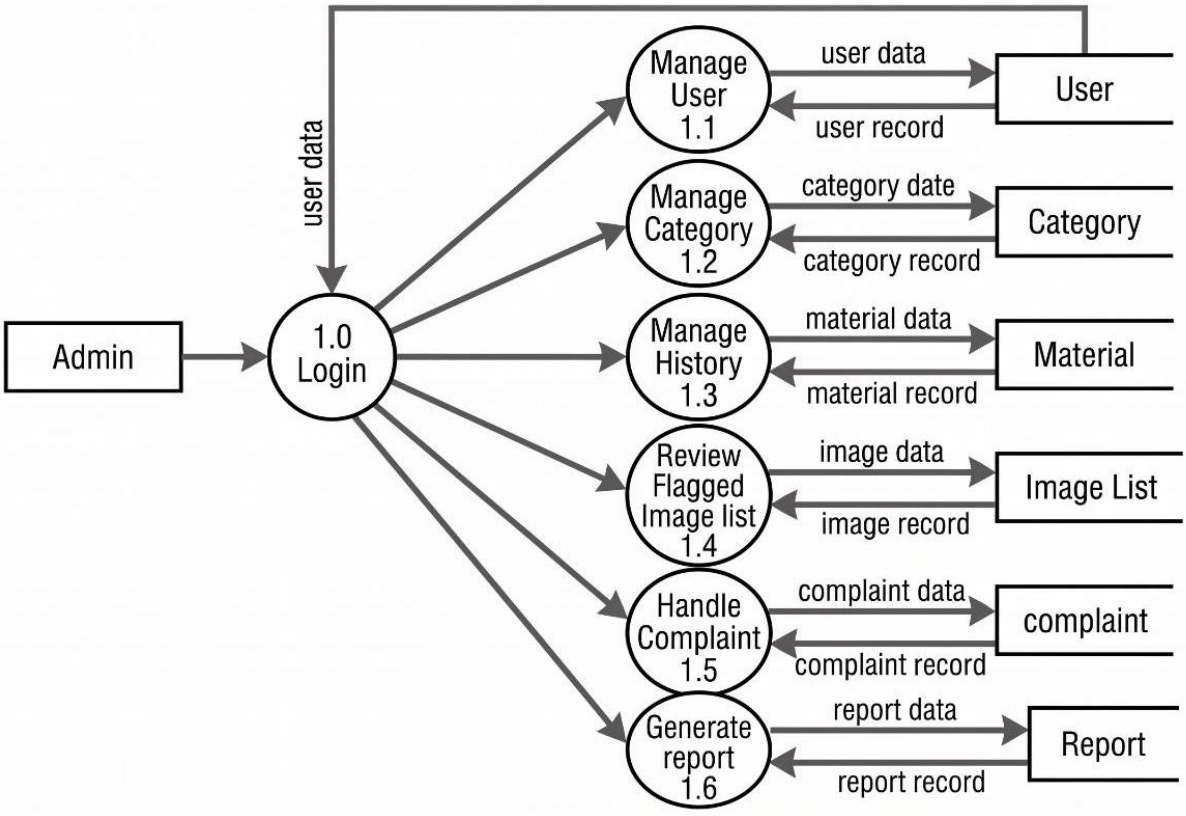


Figure 5.4: DFD Level 1 - Admi

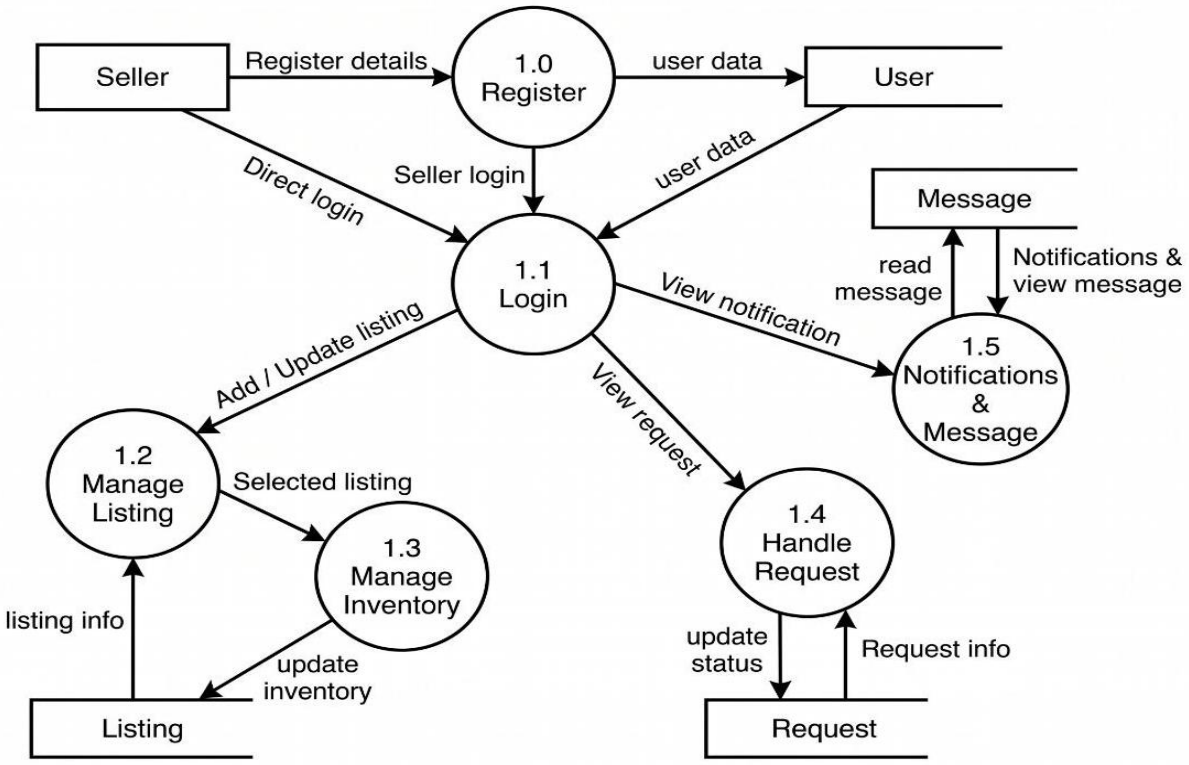


Figure 5.5: DFD Level 1 – Seller

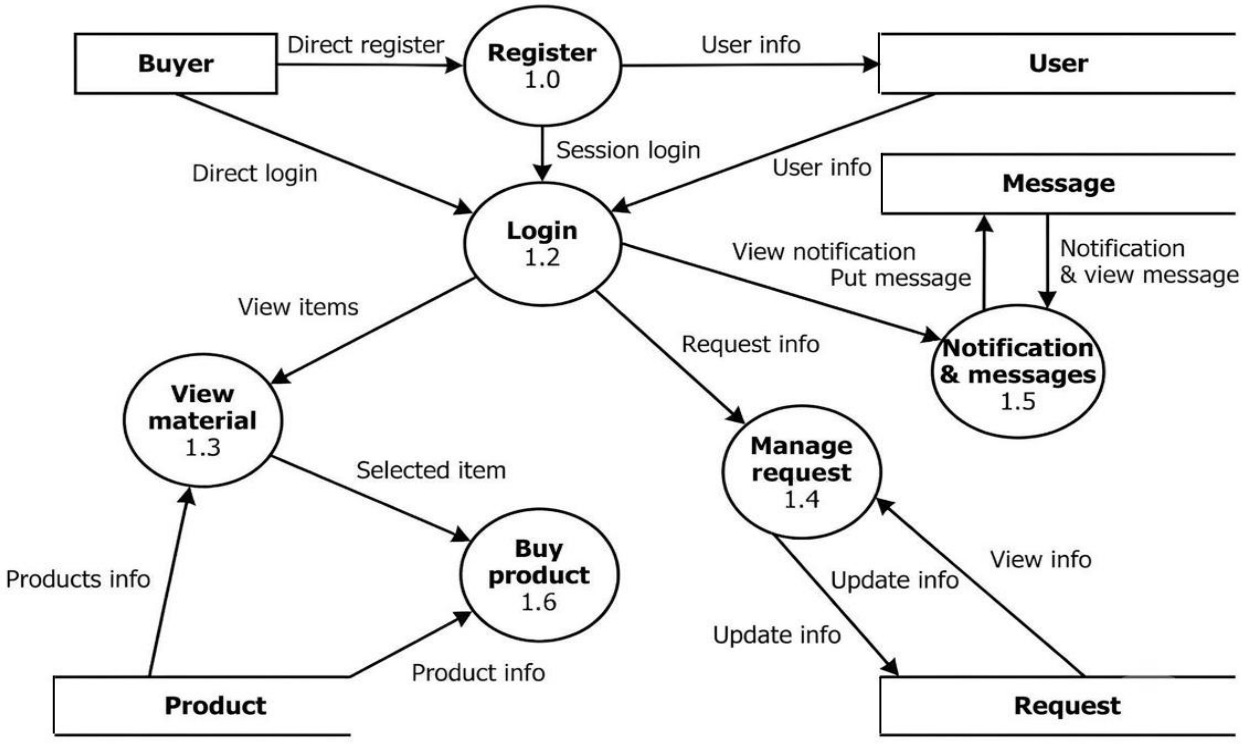


Figure 5.6: DFD Level 1 – Buyer

5.6 Module Description

Build Exchange is designed as a user-friendly web application that enables users to exchange unused and surplus construction materials through a centralized platform. It operates through three primary modules: **Buyer Module, Seller Module, and Admin Module**, which work together to provide an organized marketplace with AI-assisted image similarity detection and efficient material request management.

5.6.1 Admin Module

The Admin Module is responsible for managing and monitoring the overall operation of the Build Exchange platform. The administrator can manage registered users, monitor material listings, manage material categories, handle complaints, and view system reports. The module also allows the administrator to review images flagged by the image analysis component for potential duplicate or visually similar images. These functionalities help maintain the reliability, quality, and proper functioning of the platform.

Key functionalities:

- User management
- Material listing management
- Category management
- Complaint management
- Report generation and monitoring
- Review of flagged images
- Monitoring platform activities

5.6.2 Seller Module

The Seller Module enables users to list and manage unused or surplus construction materials available for exchange. Sellers can add material details such as name, description, category, condition, quantity, expected price, location, and images. They can edit or remove listings, view buyer requests, accept or reject requests, and coordinate pickup arrangements with buyers. The module also updates the available quantity when an exchange is completed, ensuring that the displayed inventory remains accurate.

Key functionalities:

- Seller registration and login

- Create, update, and manage material listings
- Upload material images
- Set material quantity, condition, location, and expected price
- View and respond to buyer requests
- Coordinate pickup date, time, and location
- Manage exchange status
- Update material quantity after completed collection
- View notifications and request updates

5.6.3 Buyer Module

The Buyer Module allows users to find and request unused construction materials listed on the platform. Buyers can browse available materials, search and filter listings by relevant criteria, view material details, and save materials to their favorites. They can submit requests for required materials, track request status, and coordinate pickup arrangements with sellers. This module helps buyers discover reusable construction materials while supporting the reduction of construction waste.

Key functionalities:

- Buyer registration and login
- Browse available construction materials
- Search and filter material listings
- View material details, images, quantity, and location
- Add materials to favorites
- Submit material requests
- Track request status and updates
- Communicate with sellers regarding pickup arrangements
- View notifications and exchange history

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 Implementation Plan

6.1.1 Setup and Environment Preparation

The first phase involves preparing the development environment and configuring the technologies required for the project. The MERN stack is selected for developing the web application.

- Install and configure Node.js, npm, and the required development tools.
- Create the React.js frontend and Node.js/Express.js backend projects.
- Configure MongoDB for storing user details, material listings, categories, favorites, and exchange requests.
- Configure Cloudinary for uploading and storing material images.
- Install the necessary dependencies and organize the project into appropriate folders and modules.
- Configure environment variables for database connections, cloud storage credentials, and application settings.

6.1.2 Backend Implementation

The backend phase focuses on developing the server-side logic, REST APIs, database operations, and application functionalities using Node.js, Express.js, and MongoDB.

- Develop REST APIs for user registration, login, and role-based access.
- Implement separate functionalities for buyers, sellers, and administrators.
- Create APIs for adding, viewing, updating, and deleting material listings.
- Implement search and filtering based on material category, condition, location, quantity, and expected price.
- Develop exchange request management, including request submission, acceptance, rejection, and status tracking.
- Implement inventory updates when material collection is completed.
- Integrate Cloudinary for material image uploads and retrieval.
- Implement perceptual hashing (pHash) to identify and flag potentially duplicate or visually

similar listing images.

- Test the APIs to verify correct data processing and communication between the frontend and database.

6.1.3 Frontend Implementation

The frontend phase focuses on developing an interactive and user-friendly interface using React.js, HTML, CSS, and JavaScript.

- Design a responsive user interface for different screen sizes.
- Develop registration and login pages.
- Create separate dashboards for buyers, sellers, and administrators.
- Implement material listing forms with fields for description, category, condition, quantity, expected price, location, and images.
- Develop material browsing, searching, filtering, and detailed view pages.
- Create interfaces for submitting and managing exchange requests.
- Implement favorites, request status tracking, and pickup coordination features.
- Develop administrative interfaces for managing users, listings, categories, complaints, and flagged images.
- Connect the frontend components to backend REST APIs and test the complete user workflow.

6.1.3 Security and Data Handling

The security and data-handling phase ensure that user information, material listings, and application resources are managed appropriately.

- Implement authentication to verify user identities.
- Apply role-based authorization to restrict access to buyer, seller, and administrator functions.
- Validate user inputs on both the frontend and backend to reduce invalid or harmful data.
- Store passwords securely using a suitable password-hashing algorithm.
- Use environment variables to protect database credentials, API keys, and Cloudinary secrets.
- Apply appropriate access controls to database operations and material listings.
- Validate uploaded image files by checking their file types and sizes.

- Handle API errors and database failures appropriately without exposing sensitive information.
- Ensure that material quantities and exchange request statuses are updated consistently.
- Test the application for common security issues and verify that users can access only the functions permitted by their roles.

6.2 Testing and Various Types of Testing Used

6.2.1 Unit Testing

Unit testing verifies individual components or functions of the application independently. It helps identify errors in specific modules before they are integrated into the complete system.

No.	Procedures	Expected Result	Actual Result	Pass/Fail
1	Test user registration with valid details.	User details should be accepted and saved successfully.	The user was registered and the details were saved successfully.	Pass
2	Test login with valid credentials.	The user should be authenticated successfully.	The user was authenticated and redirected to the appropriate dashboard.	Pass
3	Test material listing creation with valid details.	The material listing should be created successfully.	The material listing was created and saved successfully.	Pass
4	Test material quantity update.	The updated quantity should be saved correctly.	The material quantity was updated correctly.	Pass
5	Test the pHash function using identical images.	The system should identify the images as potential duplicates.	The identical images produced matching perceptual hashes and were identified as potential duplicates.	Pass

Table 6.1 Unit test case and results

6.2.2 Integration Testing

Integration testing verifies whether different modules and components communicate and function correctly when connected. It focuses on interactions between the frontend, backend APIs, database, cloud image storage, and image analysis module.

No.	Procedures	Expected Result	Actual Result	Pass/Fail
1	Submit registration details through the frontend.	The backend should process the request and store the user details in MongoDB.	The registration request was processed, and the user details were stored in MongoDB.	Pass
2	Add a material listing with an image.	The listing details should be saved in MongoDB and the image should be uploaded to Cloudinary.	The listing details were saved, and the image was uploaded to Cloudinary.	Pass
3	Submit an exchange request as a buyer.	The request should be stored and made available to the seller.	The exchange request was saved and displayed to the seller.	Pass
4	Accept or reject an exchange request as a seller.	The request status should be updated and displayed correctly.	The request status was updated and displayed correctly.	Pass
5	Upload an image for similarity analysis.	The pHash module should compare the image with existing images and flag potential matches.	The image was processed, and matching or similar images were flagged for review.	Pass

Table 6.2 Integration testing and results

6.2.3. System Testing

System testing evaluates the complete Build Exchange application to ensure that all major functionalities operate together according to the project requirements.

No.	Procedures	Expected Result	Actual Result	Pass/Fail
1	Register and log in as a buyer, seller, and administrator.	Each user should access the appropriate dashboard and permitted functions.	Each role accessed its corresponding dashboard and available functions.	Pass
2	Browse and search for	Relevant material listings	The system displayed	

	construction materials using filters.	should be displayed according to the selected filters.	listings matching the selected search criteria.	Pass
3	Complete the exchange request workflow from submission to pick up coordination.	The buyer and seller should be able to manage the request and coordinate collection.	The request workflow operated correctly, allowing users to manage requests and coordinate pickup.	Pass
4	Complete a material collection and update the listing quantity.	The available quantity should be reduced by the collected amount, according to the implemented workflow.	The available quantity was reduced according to the completed collection.	Pass
5	Attempt to access administrator functions using a buyer account.	Unauthorized access should be denied.	The buyer account was prevented from accessing restricted administrator functions.	Pass
6	Upload an image that is identical or visually similar to an existing listing image.	The system should flag potential image similarity for review.	The image analysis module flagged the potential match for administrator review.	Pass
7	Test the application using different screen sizes and common user workflows.	The interface should remain usable and the main functions should work correctly.	The interface remained usable, and the tested workflows operated as expected.	Pass

Table 6.3 System testing and result

CHAPTER 7

RESULTS AND DISCUSSION

7.1 Advantages

- **Centralized Marketplace:** Provides a single platform for buyers and sellers to exchange construction materials.
- **Easy Material Discovery:** Enables users to search and filter materials based on category, location, condition, and availability.
- **Efficient Request Management:** Simplifies material requests, seller responses, and pickup coordination.
- **Inventory Management:** Helps sellers maintain updated material quantities after completed collection.
- **AI-Assisted Image Analysis:** Uses perceptual hashing (pHash) to identify potential duplicate or visually similar images for review.
- **Cloud Image Storage:** Uses Cloudinary to store and retrieve material images.
- **Role-Based Access:** Provides separate functionalities for buyers, sellers, and administrators.
- **Cost-Effective Reuse:** Helps users find reusable materials and potentially reduce material procurement costs.
- **Improved Resource Utilization:** Ensures that unused construction materials can be put to practical use instead of being wasted.
- **Improved Communication:** Helps buyers and sellers coordinate material requests and pickup arrangements.
- **Time Saving:** Reduces the time required to find suitable surplus construction materials.
- **User-Friendly Interface:** Provides separate interfaces and functions for buyers, sellers, and administrators according to their roles.

7.2 Limitations

- **Internet Dependency:** Users need an internet connection to access the platform.
- **Image Analysis Accuracy:** Perceptual hashing may not identify every duplicate and can flag visually similar but different images.
- **Limited Fraud Detection:** Image similarity analysis alone cannot confirm whether an image is fake or AI-generated.
- **User-Provided Information:** The accuracy of material details depends on the information supplied by sellers.
- **Location Accuracy:** Pickup coordination depends on the accuracy of the location details provided by users.
- **No Online Payment:** The platform does not support online payments or financial transactions.
- **Material Availability:** Listings may become outdated if sellers do not update material quantities or availability.
- **Cloud Storage Dependency:** Cloudinary-based image services depend on service availability and account limits.
- **Manual Pickup Coordination:** Buyers and sellers must coordinate collection arrangements themselves.
- **Limited Physical Verification:** The platform cannot guarantee the actual quality or condition of materials without physical inspection.

7.3 Screenshots

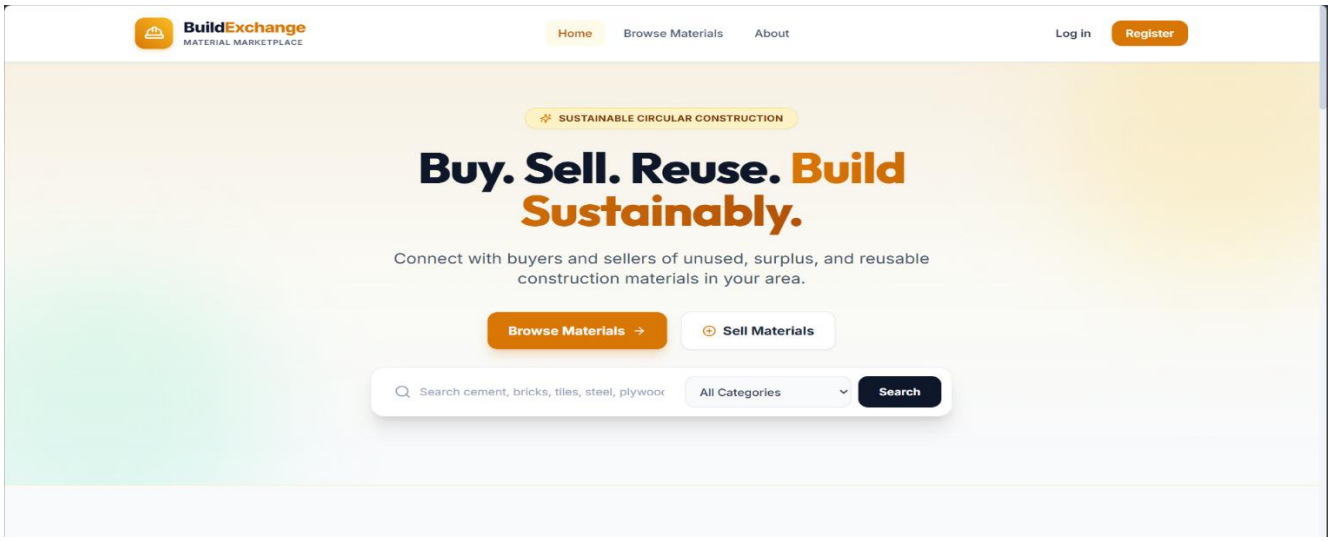


Figure 7.1: Home Page

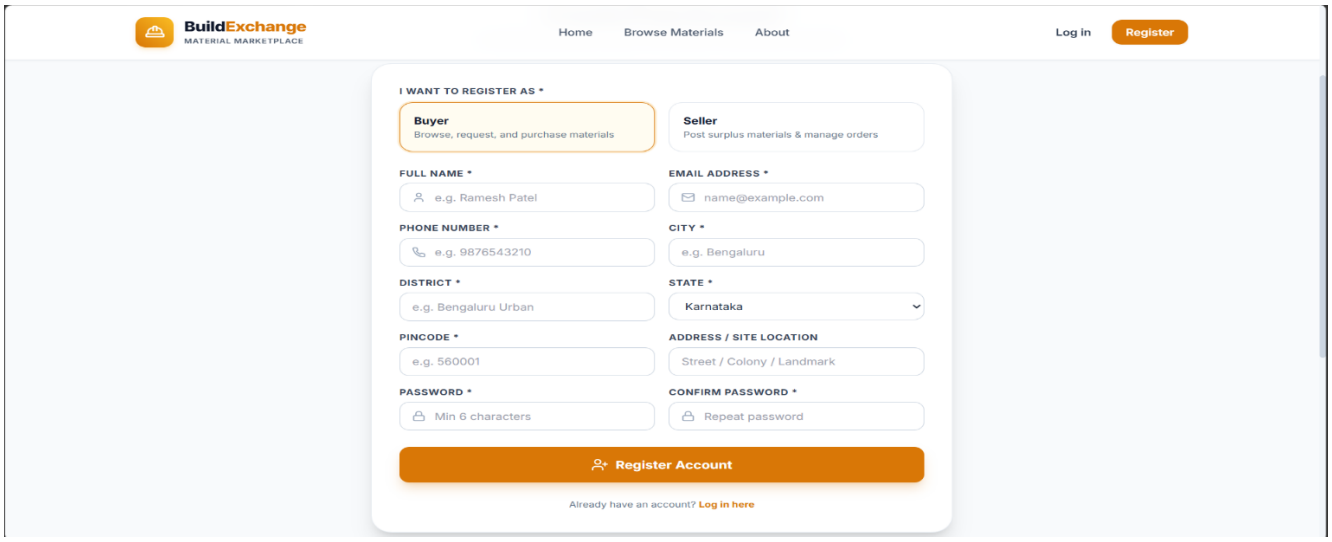


Figure 7.2: Registration Page

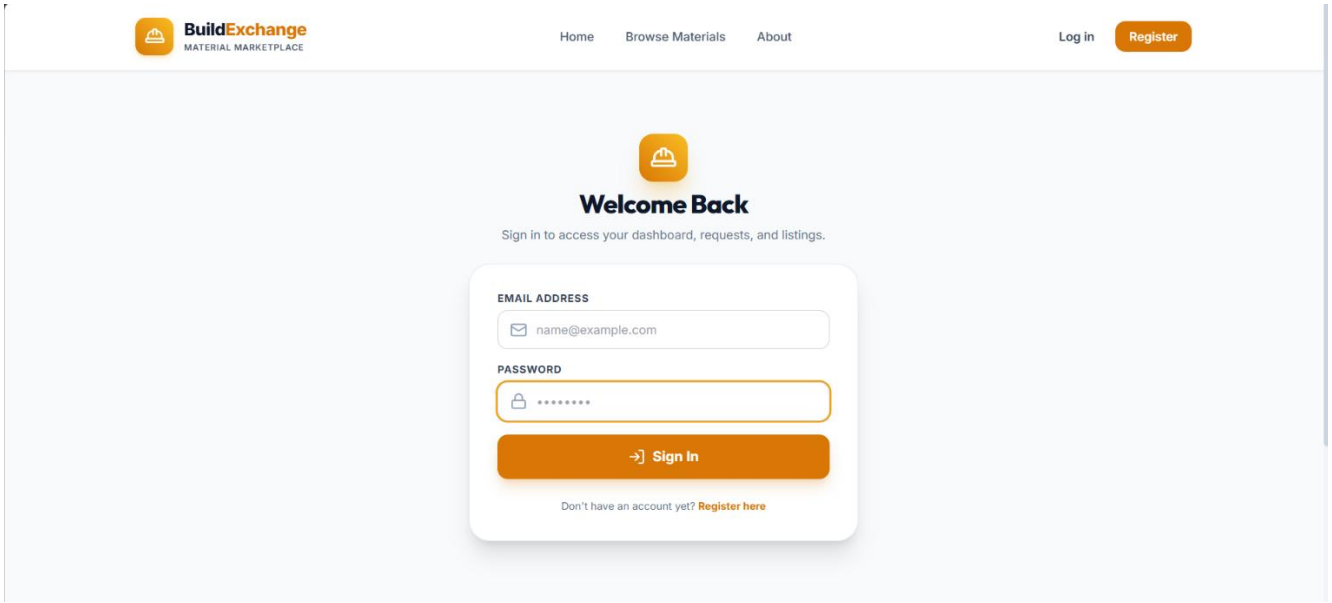


Figure 7.3: Login Page

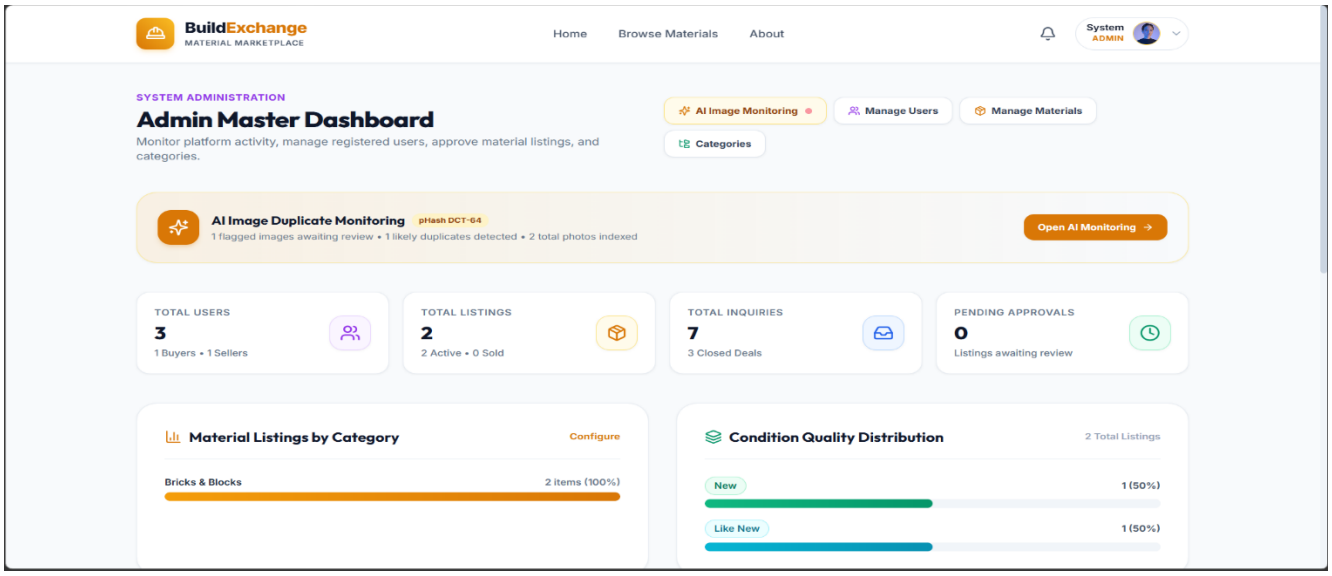


Figure 7.4: Admin Dashboard

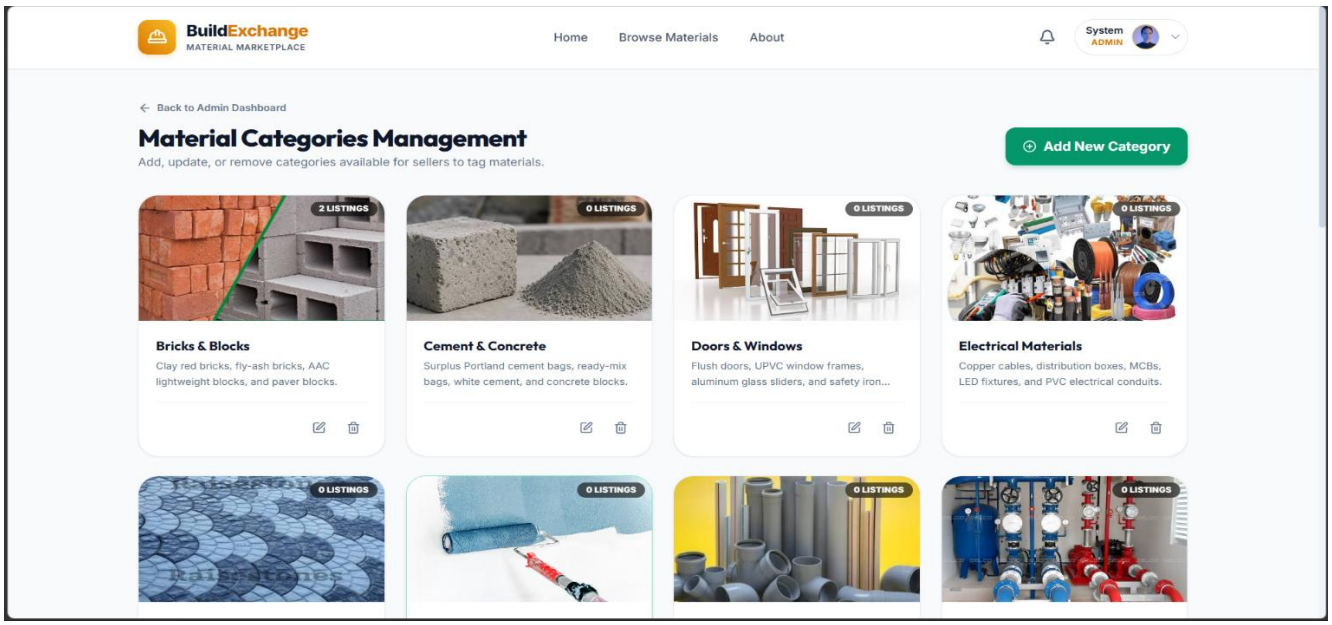


Figure 7.5: Admin Material Management

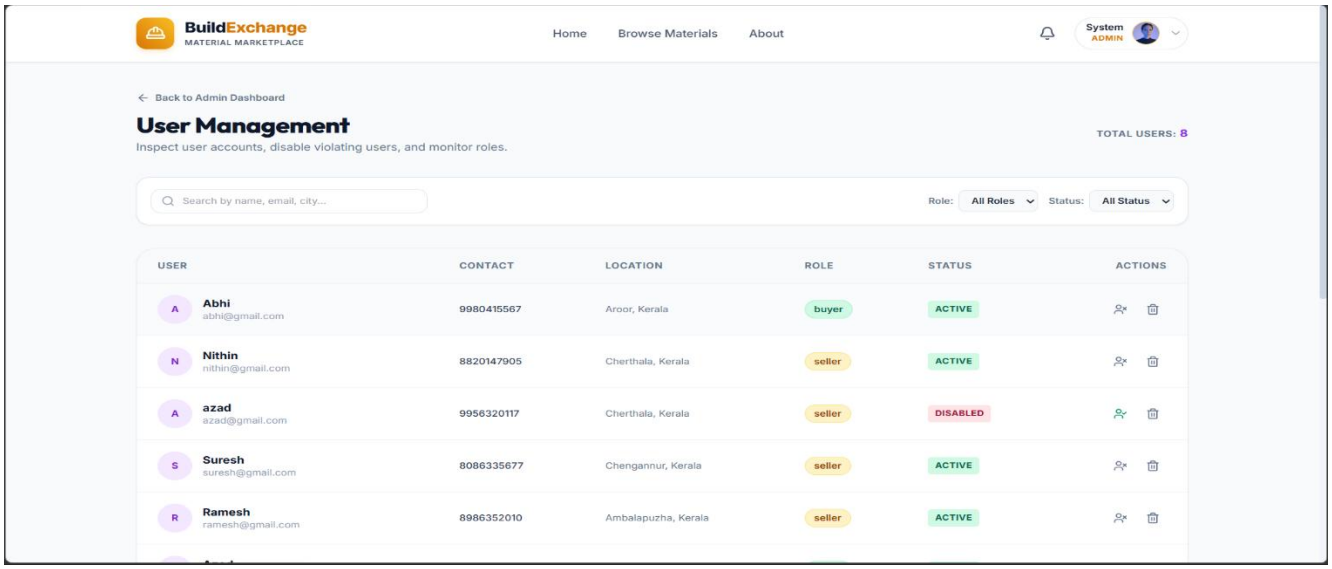


Figure 7.6: Admin User Management

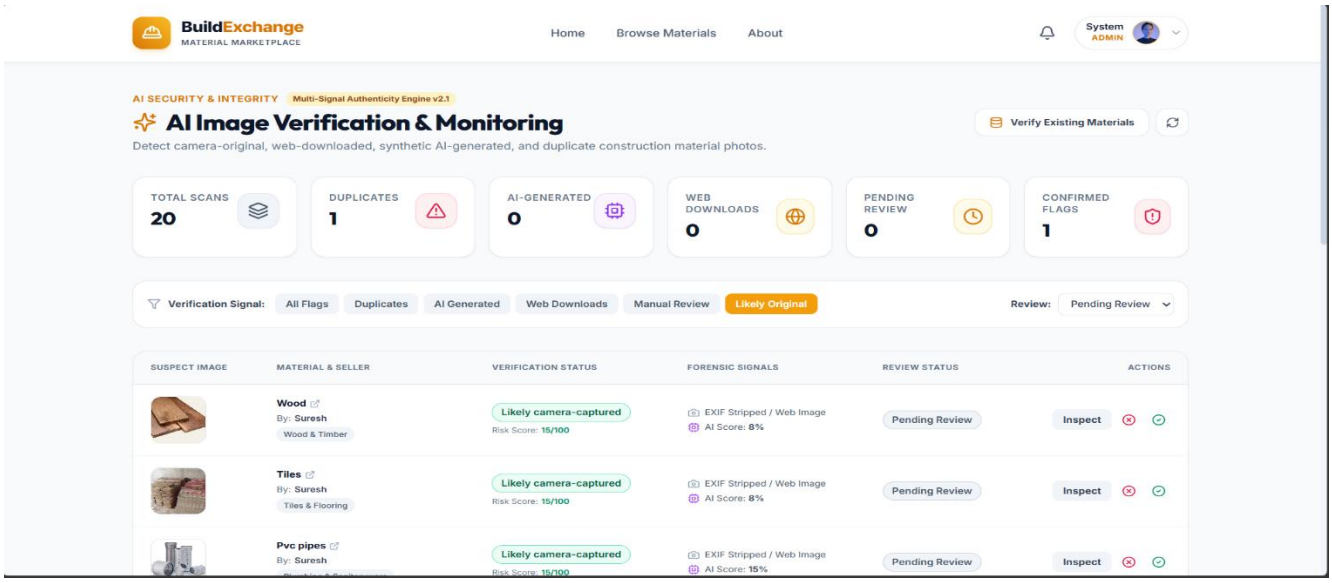


Figure 7.7: AI Image Monitoring

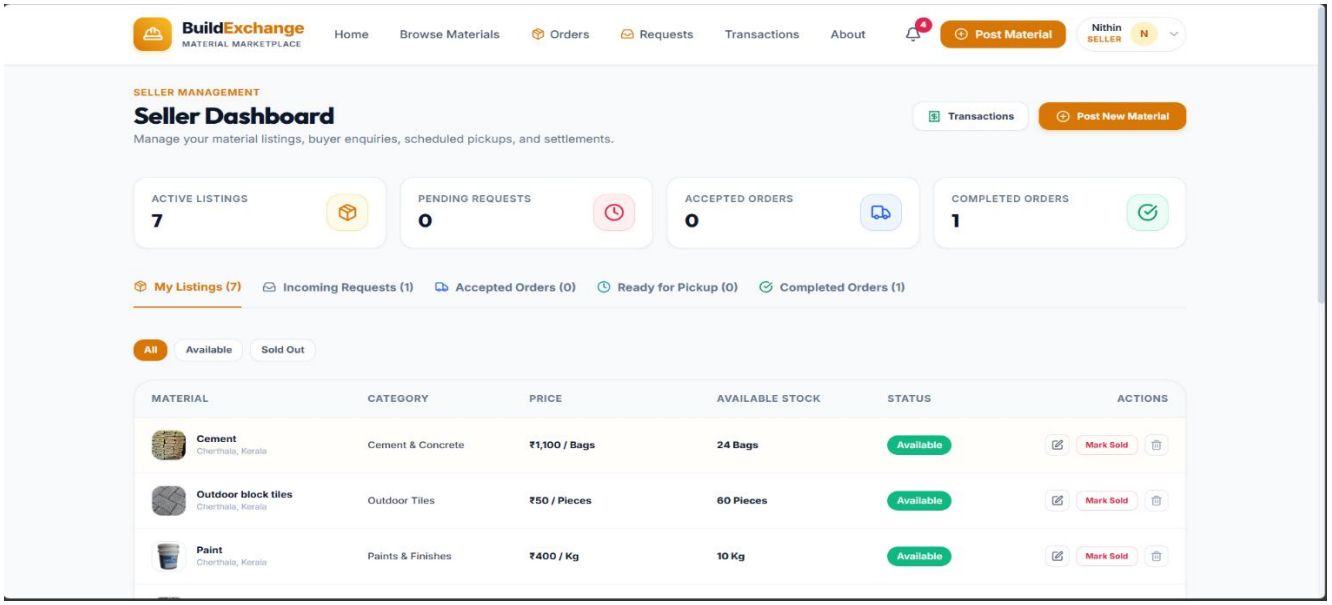


Figure 7.8: Seller Dashboard

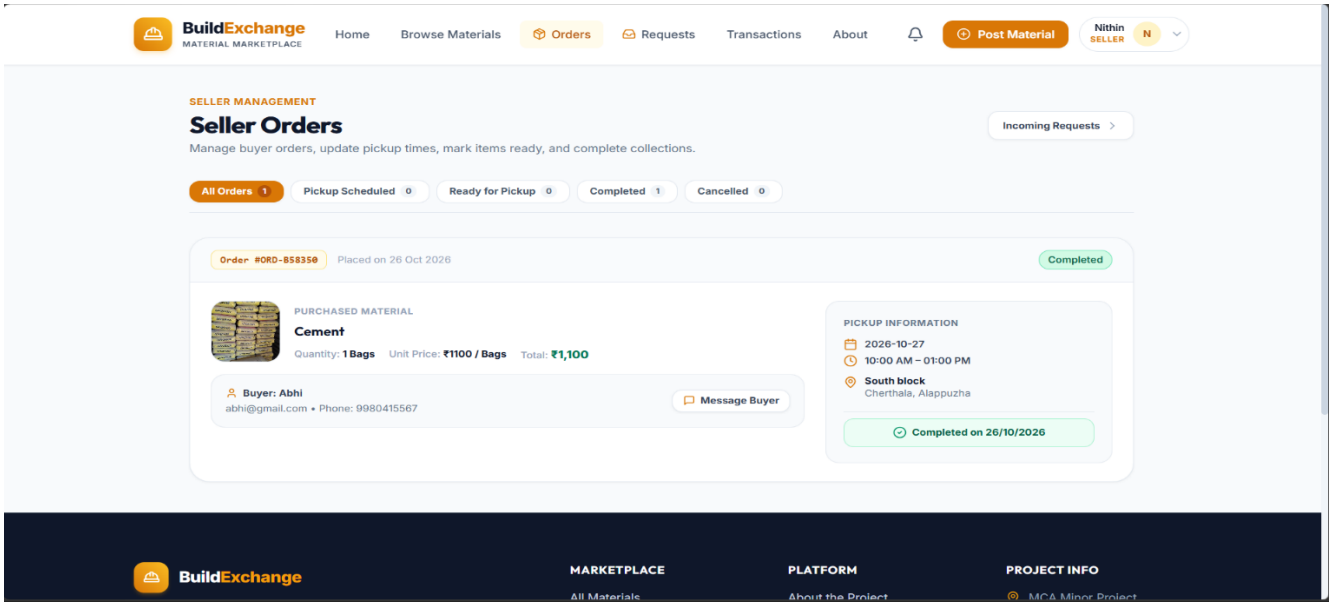


Figure 7.9: Seller Order

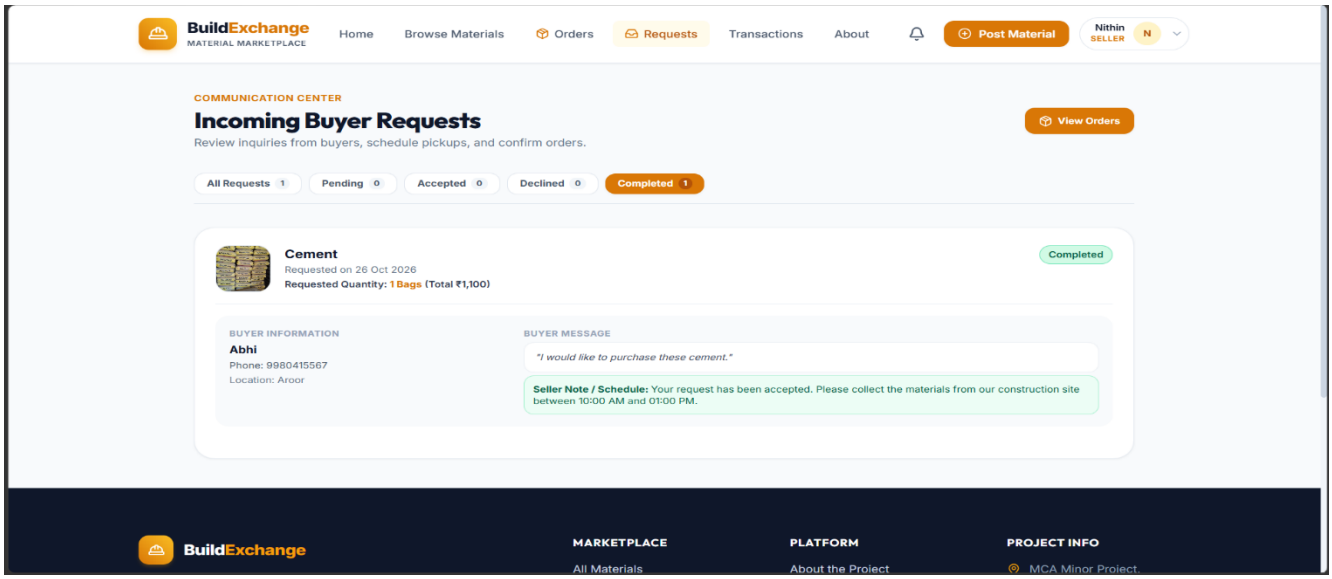


Figure 7.10: Seller Request

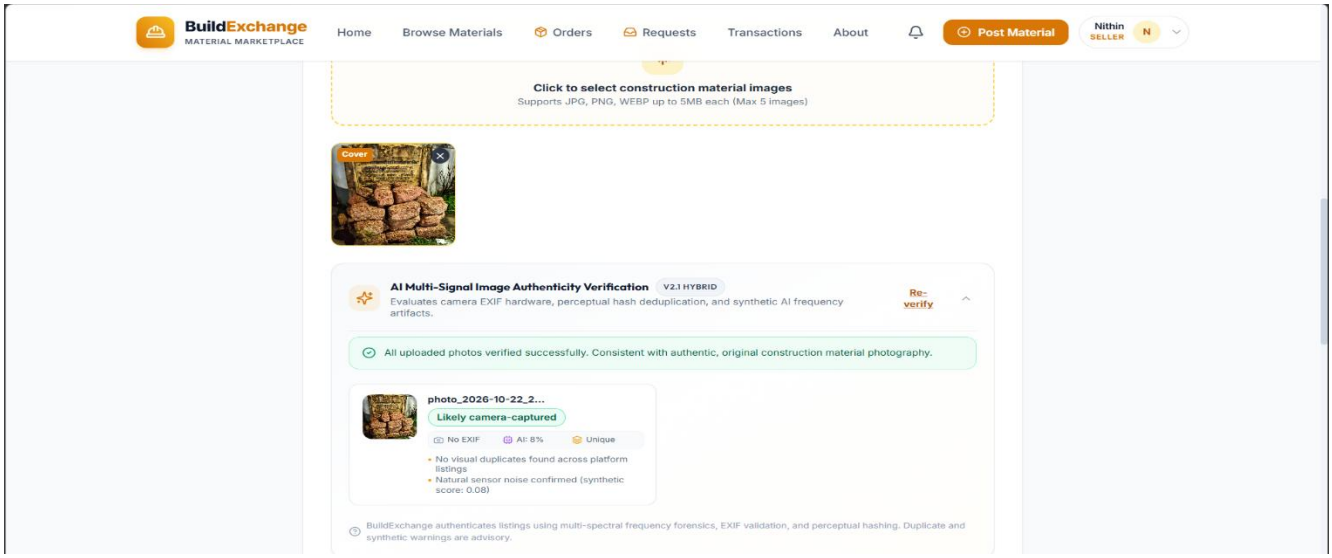


Figure 7.11: Image Verification

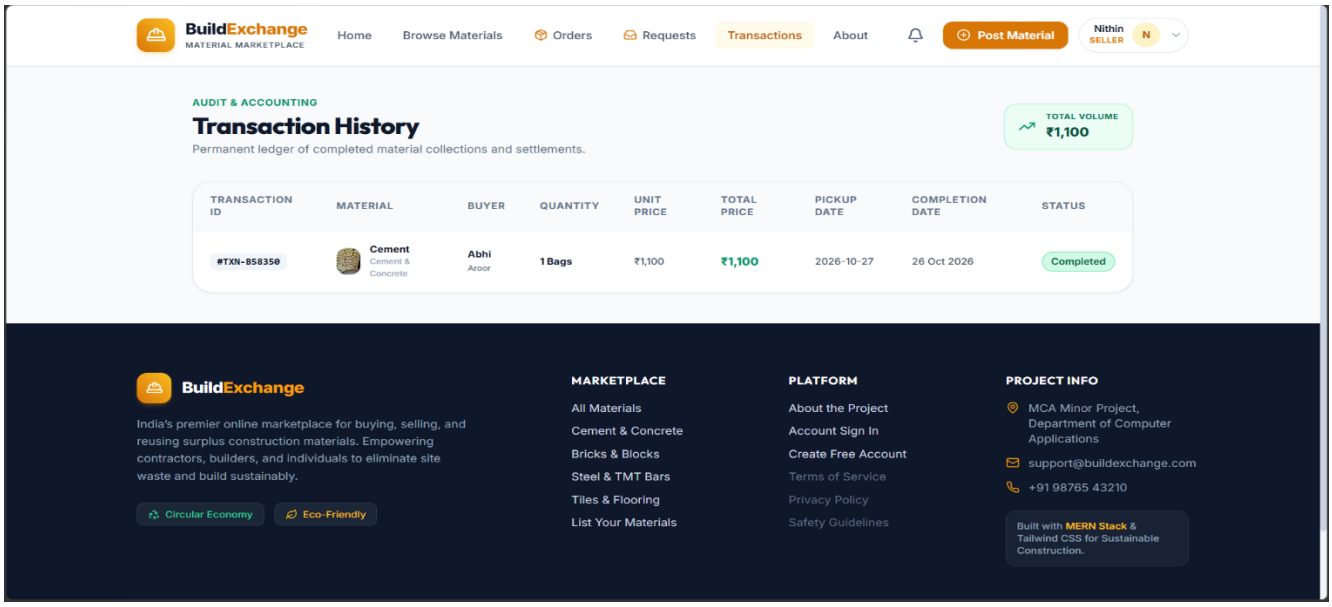


Figure 7.12: Sellers History

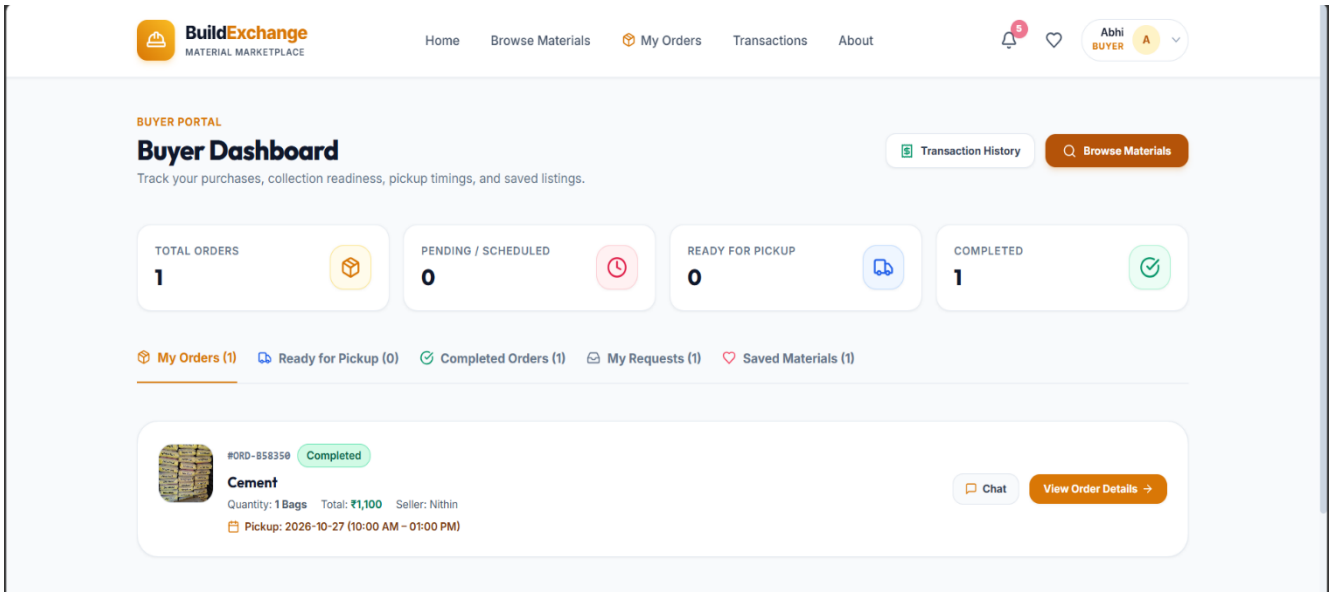


Figure 7.13: Buyer Dashboard

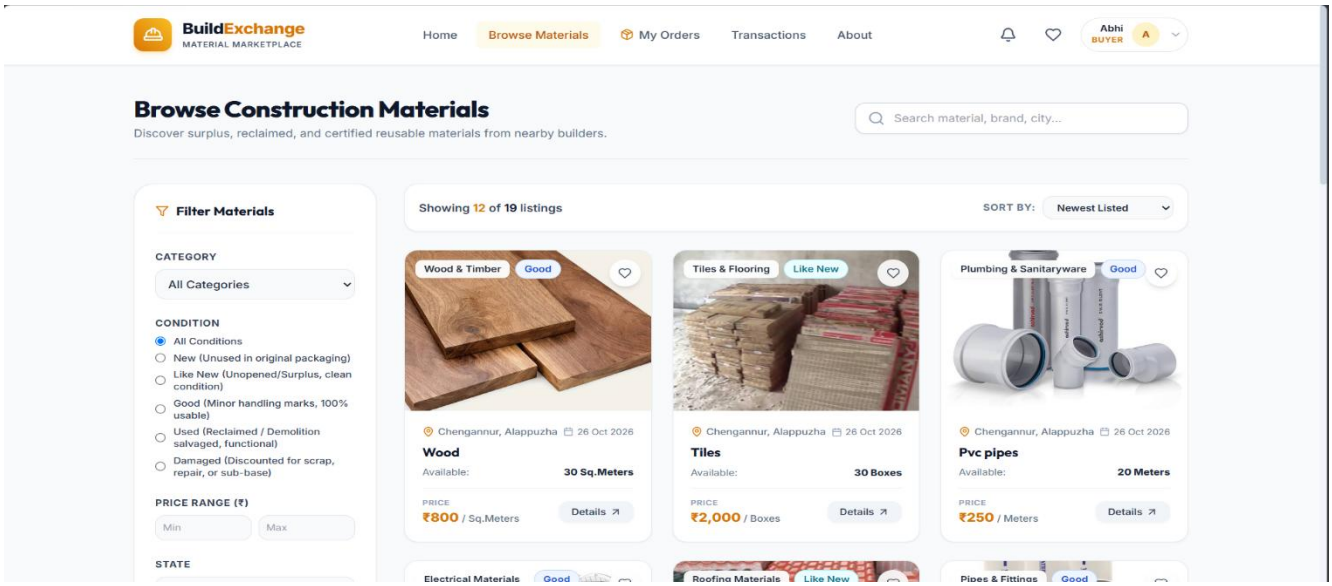


Figure 7.14: Browse Materials

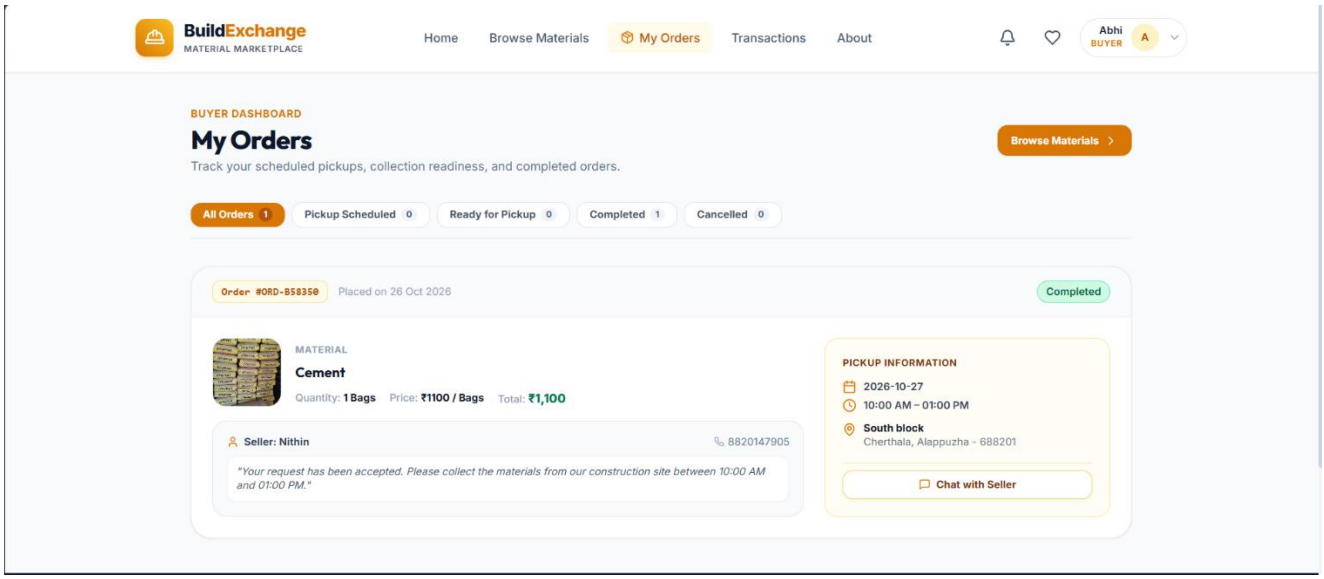


Figure 7.15: Buyer Order

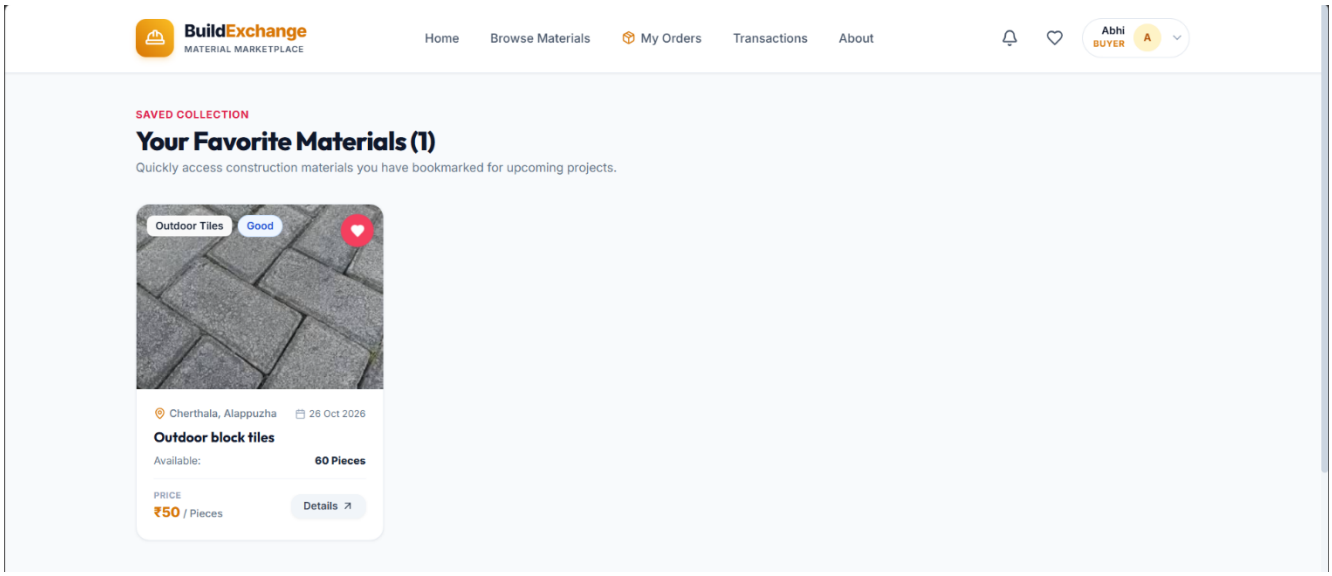


Figure 7.16: Favorite

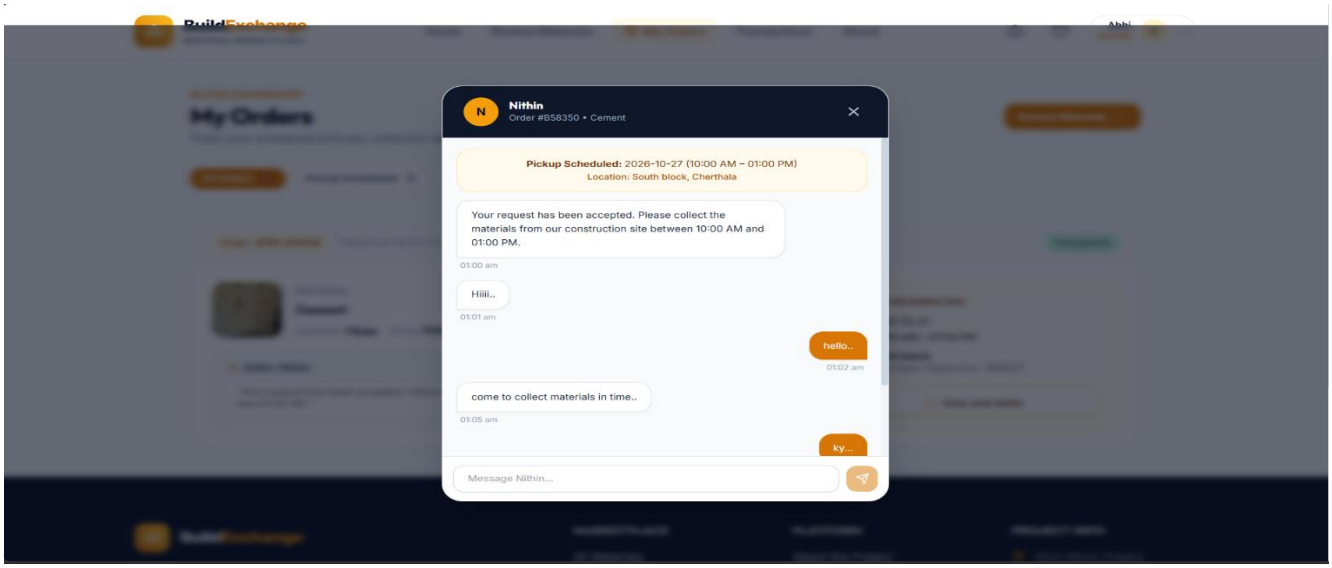


Figure 7.17: Chat

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

Build Exchange is an AI-powered construction material exchange platform developed using the MERN stack to promote the reuse of unused and surplus construction materials. The system provides a centralized platform where sellers can list available materials and buyers can search, view details, and submit exchange requests. It simplifies material listing, request management, inventory updates, and pickup coordination through separate modules for buyers, sellers, and administrators.

The integration of MongoDB for data management and Cloudinary for image storage supports efficient handling of material information and images. The perceptual hashing (pHash) technique assists in identifying potentially duplicate or visually similar listing images, helping administrators review suspicious listings. Although the system does not support online payments and relies on user-provided information and physical material verification, it offers a practical approach to improving the exchange of surplus construction materials.

In conclusion, Build Exchange aims to reduce construction waste, improve resource utilization, and encourage sustainable construction practices. The platform can be further enhanced in the future with advanced image analysis, improved location-based services, and additional features to make material exchange more efficient and reliable.

8.2 Future Scope

Build Exchange has considerable potential for further development to improve the efficiency, reliability, and usability of construction material exchange. The following enhancements can be introduced in future versions of the system.

- Advanced AI Image Detection: The image analysis module can be enhanced to better identify duplicate images, manipulated images, and AI-generated images using advanced machine learning techniques.
- Improved Location-Based Search: GPS integration and map-based search can help buyers identify available materials near their location and reduce transportation difficulties.

- Mobile Application Development: A dedicated Android and iOS application can make the platform more accessible and convenient for users.
- Real-Time Notifications: Email, SMS, or push notifications can inform users about new requests, request status changes, and pickup arrangements.
- Enhanced Search and Filtering: Advanced filtering and recommendation features can help buyers find suitable materials based on category, condition, quantity, expected price, and distance.
- User Ratings and Reviews: A feedback system can help users evaluate their exchange experiences and improve trust between buyers and sellers.
- Advanced Analytics and Reports: Data analytics can help administrators monitor material reuse, user activity, popular categories, and the platform's contribution to waste reduction.
- Environmental Impact Estimation: Future versions can estimate the amount of construction waste diverted from disposal and the environmental benefits associated with material reuse.
- Multilingual Support: Support for regional languages can make the platform more accessible to users from different backgrounds.
- Scalability and Wider Adoption: The platform can be expanded to cover more locations, construction material categories, suppliers, and recycling organizations.

These enhancements can transform Build Exchange into a more intelligent, accessible, and sustainable platform for construction material reuse. By continuously improving its features and capabilities, the system can contribute to reducing construction waste, conserving resources, and promoting a circular economy in the construction industry.

REFERENCES

[1] C. De Wolf, B. S. Byers, D. Raghu, M. Gordon, V. Schwarzkopf, and E. Triantafyllidis, “D5 digital circular workflow: Five digital steps towards matchmaking for material reuse in construction,” *npj Materials Sustainability*, vol. 2, Art. no. 36, 2024.

[2] T. Menny, S. Le Guirriec, and C. De Wolf, “The butterfly matchmaking model for circular construction: Towards a digital matchmaking platform tailored to French policy,” *Sustainable Production and Consumption*, vol. 49, pp. 130–143, 2024.

[3] A. Olumo and C. Haas, “Building material reuse: An optimization framework for sourcing new and reclaimed building materials,” *Journal of Cleaner Production*, vol. 479, Art. no. 143892, 2024.

[4] D. Sirimewan, A. F. Kineber, S. Raman, R. Garcia, and M. Arashpour, “Leveraging large-scale vision foundation models for automated construction and demolition waste recognition,” *Advanced Engineering Informatics*, vol. 68, pt. A, Art. no. 103593, 2025.

[5] S. Heo and S. Na, “Developing WasteSAM: A novel approach for accurate construction waste image segmentation to facilitate efficient recycling,” *Waste Management & Research*, vol. 43, no. 7, pp. 1048–1059, 2025.

[6] Y. Jakhar and M. D. Borah, “Effective near-duplicate image detection using perceptual hashing and deep learning,” *Information Processing & Management*, vol. 62, no. 4, Art. no. 104086, 2025.